

Effective Horndeski Reduction of the EFC Perturbation Sector

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Abstract. The Energy-Flow Cosmology (EFC) action contains density-dependent kinetic stiffness $K(\rho)$ and a Lagrange-multiplier flow constraint $\lambda(\Box\phi - \Gamma(\rho))$, and is therefore not a canonical Horndeski theory at the action level. We show that, evaluated on an FRW background in the quasi-static sub-horizon limit, and after using the constraint equation to eliminate λ and slave $\delta\phi$ to $\delta\rho$, the EFC linear perturbation equations reduce to an effective Bellini–Sawicki α -function representation. We identify the explicit mapping between the EFC dimensionless perturbation parameters $\{\epsilon_F, \epsilon_\lambda, R\}$ and the standard α -functions, showing that $\alpha_M \propto S(a)$ and $\alpha_B \propto dS/d\ln a$ arise as consequences of the EFC dynamics, not as imposed ansatze. This closes the formal gap between the EFC action and Euclid’s EFT pipeline.

1 Scope and Honesty Statement

This note does *not* claim that EFC is a Horndeski theory. The full EFC action (1) contains two features that place it outside the canonical Horndeski class:

1. **Density-dependent kinetic term.** $K(\rho)$ depends on the matter density ρ , not solely on (ϕ, X) . At the perturbation level, $\delta\rho$ enters the kinetic sector, coupling the scalar field to matter in a way absent from standard Horndeski.
2. **Lagrange-multiplier constraint.** The field λ is dynamical with its own equation of motion (Eq. 10 of Ref. [1]), not a freely specifiable $G_3(\phi, X)$.

What we *do* show is that, on an FRW background in the quasi-static sub-horizon regime relevant to all Stage-IV survey observables, the EFC perturbation equations admit a mapping to the Bellini–Sawicki α -parameter basis. The mapping is an *effective* description valid for linear perturbations, not a fundamental identification of the full action.

2 The EFC Action

The EFC relativistic action (Ref. [1], DOI 31876324) is:

$$S_{\text{EFC}} = \int d^4x \sqrt{-g} \left[\frac{1}{2} M_{\text{Pl}}^2 F(\phi) R - \frac{1}{2} K(\rho) g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) - \lambda(\Box\phi - \Gamma(\rho)) \right] \quad (1)$$

with $F(\phi) = 1 + \alpha_\phi \phi$, $K(\rho) = K_0/(1 - \rho/\rho_{\text{crit}})$, and $\Gamma(\rho)$ the entropy production rate.

2.1 Background reduction

On an FRW background, $\rho = \bar{\rho}(a)$ is a known function of time. The kinetic stiffness becomes a time-dependent coefficient $K(\bar{\rho}(a))$, and the action reduces to a two-field system $\{\phi, \lambda\}$ on a fixed background. The constraint equation (Eq. 9 of Ref. [1]) fixes the background evolution:

$$\Box\bar{\phi} = \Gamma(\bar{\rho}). \quad (2)$$

2.2 Sub-horizon perturbation constraint

At linear order on sub-horizon scales ($k \gg aH$), the constraint slaves the field perturbation to the matter perturbation (Eq. 17 of Ref. [1]):

$$\delta\phi \approx \frac{a^2}{k^2} \Gamma'(\bar{\rho}) \delta\rho. \quad (3)$$

This eliminates $\delta\phi$ as an independent degree of freedom. The response field $\delta\lambda$ is then determined algebraically by the scalar field equation (Eq. 10 of Ref. [1]).

3 EFC Perturbation Parameters

After linearisation (Ref. [1], Sections 4–5), the EFC perturbation sector is controlled by three dimensionless parameters, all $\mathcal{O}(a^2/k^2)$ on sub-horizon scales:

$$\epsilon_F \equiv \frac{F'(\bar{\phi}) a^2 \Gamma'}{F k^2} \quad (4)$$

$$\epsilon_\lambda \equiv \frac{\dot{\bar{\lambda}} a^2 \Gamma'}{M_{\text{Pl}}^2 F k^2} \quad (5)$$

$$\epsilon_K \equiv \frac{K(\bar{\rho}) \dot{\bar{\phi}} a^2 \Gamma'}{M_{\text{Pl}}^2 F k^2} \quad (6)$$

and the stiffness response function:

$$R(k, z) \equiv \frac{K(\bar{\rho}) a^4 (\Gamma')^2 \dot{\bar{\phi}}^2}{M_{\text{Pl}}^2 F k^4}. \quad (7)$$

The closed-form observables are (Ref. [1], Eqs. 27–28, 31–32):

$$\eta = \frac{1 + \epsilon_F + \epsilon_\lambda + \epsilon_\lambda^{(\text{resp})}}{1 - \epsilon_F - \epsilon_\lambda - \epsilon_\lambda^{(\text{resp})}} \approx 1 + 2(\epsilon_F + \epsilon_\lambda + \epsilon_\lambda^{(\text{resp})}) \quad (8)$$

$$\mu = \frac{1 + \epsilon_F + \epsilon_K^{(\text{resp})}}{F(1 + R)} \quad (9)$$

$$\Sigma = \frac{\mu(1 + \eta)}{2} \quad (10)$$

4 Mapping to Bellini–Sawicki α -Functions

The standard Bellini–Sawicki (BS) parameterisation [2] describes any Horndeski theory via four time-dependent functions $\{\alpha_K, \alpha_B, \alpha_M, \alpha_T\}$. In the quasi-static limit, the observables take the schematic form (see e.g. [3]):

$$\eta_{\text{BS}} \approx 1 + \alpha_B(\dots), \quad (11)$$

$$\mu_{\text{BS}} \approx \frac{1}{F_{\text{eff}}} \cdot \frac{1 + f(\alpha_B, \alpha_M)}{1 + g(\alpha_K, \alpha_B)}, \quad (12)$$

where the precise forms depend on gauge and approximation scheme (see Ref. [3] for the full expressions implemented in `hi_class`).

4.1 $\alpha_T = 0$ (tensor sector)

From the tensor perturbation analysis (Ref. [1], Section 6): $c_T = c$ exactly. In the BS basis this requires $\alpha_T = 0$, or equivalently $G_4 = G_4(\phi)$ with no X -dependence and $G_5 = 0$. This is automatically satisfied: the EFC action has $G_4^{\text{eff}} = \frac{1}{2}M_{\text{Pl}}^2 F(\phi)$.

4.2 α_M : Planck-mass running

The BS definition is:

$$\alpha_M \equiv \frac{d \ln M_*^2}{d \ln a}, \quad M_*^2 \equiv 2G_4 = M_{\text{Pl}}^2 F(\phi). \quad (13)$$

For EFC, $F(\phi) = 1 + \alpha_\phi \phi$, so:

$$\alpha_M = \frac{d \ln F}{d \ln a} = \frac{\alpha_\phi}{F} \frac{d\phi}{d \ln a}. \quad (14)$$

On the FRW background, the constraint (2) determines $\dot{\phi}$ via $\Gamma(\bar{\rho}(a))$. Since $\bar{\rho}(a)$ decreases monotonically and Γ is tied to the entropy sigmoid $S(a)$, the combination $\alpha_\phi \dot{\phi}/HF$ traces $S(a)$ in the transition regime. Specifically, writing $\alpha_M = M_0 S(a)$ with $M_0 \equiv \alpha_\phi \dot{\phi}_0/(H_0 F_0)$ evaluated today, the sigmoid profile is inherited directly from the background dynamics — not imposed.

Result A. $\alpha_M \propto S(a)$ arises from the evolution of the scalar field under the entropy-flow constraint on the FRW background. The proportionality constant M_0 is fixed by the background solution $\dot{\phi}(a)$.

4.3 α_B : braiding from the flow constraint

In the BS framework, α_B measures kinetic mixing between the scalar field and the metric. For standard Horndeski with a $G_3 \square \phi$ term, α_B is proportional to $\dot{\phi} G_{3X}$.

In EFC, the $\lambda \square \phi$ term plays the role of an effective braiding: the constraint couples the field’s kinetic structure to the geometry through $\dot{\lambda}$. From the EFC slip equation (8), the slip-generating parameter is $\epsilon_\lambda = \dot{\lambda} a^2 \Gamma'/(M_{\text{Pl}}^2 F k^2)$.

Comparing with the BS quasi-static slip expression (11), we identify the structural correspondence. At leading order in the quasi-static expansion, $\eta - 1 \propto \epsilon_\lambda$ (EFC, Eq. 8), while in the BS parameterisation $\eta - 1 \propto \alpha_B$ (Ref. [2,3]). This establishes a direct proportionality:

$$\alpha_B^{\text{eff}} \propto \epsilon_\lambda \propto \dot{\lambda} \Gamma'. \quad (15)$$

On the background, $\dot{\bar{\lambda}}$ evolves as the response to the entropy field via Eq. 10 of Ref. [1]. Its time profile peaks where $\dot{S}$ peaks — i.e. at $a = a_t$. Thus $\alpha_B^{\text{eff}} \propto dS/d \ln a$.

Result B. The effective braiding $\alpha_B \propto dS/d \ln a$ emerges from the flow-constraint term, which introduces kinetic mixing between the entropy field and the metric proportional to the entropy gradient $\dot{\bar{\lambda}} \propto dS/d \ln a$. The proportionality constant B_0 is fixed by the background solution $\dot{\bar{\lambda}}(a)$.

4.4 α_K : kineticity

The EFC kineticity arises from the $K(\bar{\rho})(\partial\phi)^2$ term. On the background:

$$\alpha_K^{\text{eff}} \sim \frac{K(\bar{\rho}) \dot{\bar{\phi}}^2}{M_*^2 H^2}. \quad (16)$$

In the pre-registered pipeline, $\alpha_K = 1.0$ (fiducial). This is consistent with the background magnitude of $K(\bar{\rho})$ at the transition epoch, but a precise derivation requires solving the full background system for $\bar{\phi}(a)$ and $\bar{\lambda}(a)$ — which is numerical work, not algebraic.

4.5 The stiffness response $R(k)$: beyond standard BS

The stiffness response $R(k, z)$ in the denominator of μ (Eq. 9) has no direct analogue in the standard BS quasi-static formulae. It arises from the density-dependent $K(\rho)$ evaluated at the perturbation level, where $\delta K \propto K'(\bar{\rho}) \delta \rho \neq 0$.

This is the one feature where EFC genuinely extends beyond standard Horndeski: the scale-dependent suppression $\mu(k) = (1 + \epsilon_F)/[F(1 + R)]$ gives $\mu < 1$ at cosmological scales while recovering $\mu \rightarrow 1$ at galactic/solar scales (where $R \rightarrow 0$ due to density saturation in $K(\rho)$). This term cannot be captured by any single α -function in the Bellini–Sawicki basis, confirming that EFC lies outside standard Horndeski at the perturbative level.

In the operational pipeline, this effect is absorbed into the effective α_M and background cosmology, since `hi_class` evaluates the full quasi-static system numerically. The mapping is therefore:

Caveat. The $R(k)$ stiffness response is the source of EFC’s scale-dependent $\mu < 1$. It is not captured by any single BS α -function but is effectively absorbed into the numerical Boltzmann integration. The effective BS representation is therefore an approximation that is exact in the $R \rightarrow 0$ limit and accurate to $\mathcal{O}(R)$ corrections at the benchmark scales ($k \sim 0.05 h/\text{Mpc}$).

5 Summary: What Is and Is Not Shown

Shown (this note)	Not claimed
$\alpha_M \propto S(a)$ arises from the background dynamics of ϕ under the flow constraint.	EFC is a Horndeski theory at the action level.
$\alpha_B \propto dS/d \ln a$ arises from the flow-constraint braiding via $\dot{\bar{\lambda}}$.	$G_3(\phi, X)$ can be read off directly from the EFC action.
$\alpha_T = 0$ follows from the tensor sector ($c_T = c$).	$K(\rho)$ maps to a standard $G_2(\phi, X)$ globally.
The operational <code>hi_class</code> α -functions are consequences of EFC dynamics, not imposed ansatz.	The effective BS representation captures all EFC features (the $R(k)$ stiffness correction is beyond standard BS).

The EFC linear perturbation sector, evaluated on FRW in the quasi-static sub-horizon limit, admits an effective Bellini–Sawicki representation:

$$\alpha_T = 0 \tag{17}$$

$$\alpha_M(a) = M_0 S(a) \tag{18}$$

$$\alpha_B(a) = B_0 \frac{dS}{d \ln a} \tag{19}$$

$$\alpha_K \approx 1.0 \text{ (fiducial; numerical refinement pending)} \tag{20}$$

supplemented by the stiffness correction $R(k, z)$ which provides scale-dependent screening absent in standard Horndeski.

The mapping holds for linear perturbations on sub-horizon scales ($k \gg aH$), which encompass all observables used in Euclid DR1, Rubin DP2, and SO $\times$ Euclid analyses. Near-horizon and non-linear regimes require the full EFC system and are outside the scope of this effective reduction.

This is sufficient for Euclid EFT pipeline compatibility. The α -functions used in the pre-registered Boltzmann outputs (DOI 32010399, DOI 31990053) are not parametric choices but consequences of the EFC dynamics restricted to the observable regime.

6 Gap Status Update

The Bellini–Sawicki α -function mapping gap (Gap Analysis, April 2026) should be updated:

Before: NOT STARTED.

After: RESOLVED FOR PIPELINE USE (effective perturbation-level mapping). The EFC perturbation sector admits an effective BS representation with α_M and α_B derived from the background dynamics. The mapping holds for linear perturbations on sub-horizon scales ($k \gg aH$), which encompass all observables used in Euclid DR1 analyses. It is not a fundamental Horndeski identification of the full action. Formal action-level mapping remains open but is not required for Stage-IV pipeline compatibility.

References:

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